

REPUBLIC OF YEMEN

Ministry of Education and Scientific Research

AL-HIKMA UNIVERSITY

Department of Information Technology

Sana'a



Department of Information Technology

Field Training Course Guide

Comprehensive Simulation Syllabus for Individual Full-Stack Software Development Project
Academic Year: 2026 | Course Duration: One Semester

Course Supervisor: Prof. Ibrahim Ahmed Al-Baltah

2026-2027

1. Introduction & Overview

This comprehensive field training course is structured around a rigorous, simulated real-world software enterprise environment. Rather than a passive internship, each student takes on the role of a Full-Stack Software Engineer to single-handedly design, build, test, document, and launch an enterprise-grade application. This format ensures complete exposure to the modern Software Development Life Cycle (SDLC) and forces students to encounter and solve realistic technical, architectural, and lifecycle engineering hurdles.

By participating in this individual project framework, students master crucial practical skills across all development tiers, transforming theoretical concepts that have been acquired during the previous semesters into highly marketable corporate capabilities. This project is specifically structured to simulate strict production environments, preparing candidates for professional engineering teams. ***It is beyond simulating the real internship to ensure students readiness for real project involvement with the required knowledge and skills.***

2. Project Requirements

- 1- At least three connected tables
- 2- At least one table has images
- 3- Main Dashboard
- 4- For every table (entity) You must create these views (Index, Details, Create, Delete, Edit) and perform input validation like (required field, datatype, length, email....)
- 5- Main Web page for displaying the information
- 6- You must use the following:
 - a. Bootstrap
 - b. Bootstrap Icons
 - c. Add any fonts
 - d. CSS variables
 - e. Bootstrap Modal
 - f. DataTable.js
- 7- Student must use GitHub, and create main branch and other branches, and perform merging.
- 8- For the mobile App the design of the whole app is consistence, must contain of the following:
 - a. Main page
 - b. Three others pages
 - c. Drawer
 - d. Button navigation bar
 - e. AppBar
 - f. SheetModal
 - g. Add custom font

Software Project Documentation Template

Cover Page

- Project Title
- Student
- Supervisor
- Version

Project Proposal (W2)

- Background
- Problem
- Objectives
- Scope
- Methodology
- Timeline
- Risks

Requirements

- Functional
- Non-functional

Design (W3)

- Clean Architecture/Repository Pattern
- Architecture Diagram
- Use Case Diagram
- Sequence Diagram
- Class Diagram

UI/UX (W4)

- Personas
- Wireframes
- Interactive Prototype
- Screenshots

Web API (W5)

- Project Structure
- Endpoints
- Validation

API Testing (W6)

- Swagger
- Postman
- Test Cases
- Results

ASP.NET MVC (W7)

- Modules
- Screenshots
- Business Logic

Flutter (W8)

- Architecture
- Screens
- Navigation
- API Integration

GitHub (W9)

- Repository
- Commits
- Branches

Documentation (W10)

- Installation
- User Manual
- Testing
- Conclusion
- References

Week	Component	Weight	Obtained Mark	Note
W2	Project Proposal	5%		
W3	Design software solutions using Clean Architecture or Repository Pattern (10%) + Use case diagram, sequence diagram, and class diagram	15%		
W4	UI/UX Design using XD, or Figma, and make interactive (wireframe)	5%		
W5	Web API Development using ASP.NET CORE 10.	10%		
W6	Test APIs using Swagger, Postman, or Scala and documentation	5%		
W7	ASP.NET MVC Application using ASP.NET CORE 10.	25%		
W8	Flutter Mobile Application	15%		
W9	GitHub	5%		
W10	Documentation	15%		
Total		100.00%		